

WhiteboardOS / VocaLabs

AI-Powered Multimodal Sketch-to-Code Living Prototype Engine

TRACK: MULTIMODAL AI | AUTHOR: JAGADEESH CHINTA | VERSION: 1.0

Project Title	WhiteboardOS — Multimodal Sketch-to-Code Engine
Lead Developer	Jagadeesh Chinta (Full-Stack AI Architect)
GitHub Repository	https://github.com/jagadeeshchinta/Sketch2Code
Live Production URL	https://sketch2-code-yd7c-git-master-jc-project.vercel.app
AI Evaluation Suite	https://sketch2-code-yd7c-git-master-jc-project.vercel.app/eval
Technology Stack	Next.js 15, TypeScript, Gemini Vision AI, TailwindCSS, SSE

1. What We Built & How It Works

A. The Problem

The gap between initial design ideation (drawings on paper, napkins, or whiteboards) and a coded, clickable prototype is the single largest bottleneck in frontend product development. Translating physical sketches into clean code typically requires 3 to 5 days, creates context loss, and consumes valuable engineering hours.

B. Our Solution

WhiteboardOS is an autonomous multimodal generative compiler that converts hand-drawn paper sketches, live camera scans, browser digital drawings, and spoken voice blueprints into functional, responsive, interactive web prototypes in under 10 seconds using Google Gemini Multimodal Vision AI.

C. End-to-End Pipeline Architecture

- 1. Multimodal Ingestion: Voice dictation, live webcam scanner, photo upload dropzone, and digital canvas board.
- 2. Vision AI Engine (/api/analyze): Detects 30+ UI components (navbars, cards, menus, hours, locations, buttons, forms) with spatial bounding coordinates and confidence scores.
- 3. Real-Time Code Synthesis (/api/generate/stream): Streams production HTML5, CSS, and React JSX token-by-token via Server-Sent Events (SSE) into an animated macOS terminal HUD.
- 4. Sandboxed Living Prototype & Conversational Iteration: Renders prototype in an isolated iframe with real-time natural language iteration (e.g. "Make buttons glowing amber"), version history time-travel, and 1-click ZIP export.

2. Author Contribution & Work Done

As the Lead Architect and Full-Stack AI Engineer, I designed and developed the entire end-to-end platform from scratch, building deep engineering systems beyond thin API wrappers:

- Multi-Model Cascading & Key Rotation**
Built automated failover across gemini-3.5-flash, gemini-3.6-flash, and gemini-flash-latest with multi-key rotation to eliminate HTTP 429 quota exhaustion.
- In-Browser Heuristic CV Engine**
Built an HTML5 Canvas spatial density analyzer and offline heuristic compiler that produces working prototypes even if network connectivity is 100% severed.
- AST DOM Self-Healing Guardrails**
Developed guardrails.ts to validate generated HTML, auto-repair unclosed containers, inject responsive viewport meta tags, and strip unverified scripts.
- Client-Side Image Pre-Processor**
Built an automated canvas filter that downscales 4K mobile photos to 1024px, converts them to WebP, and applies adaptive contrast stretching for faint pencil sketches.
- Real-Time SSE Streaming Pipeline**
Implemented bidirectional Server-Sent Events delivering token streams directly into a syntax-highlighted terminal.
- Liquid Glass Design System**
Crafted custom CSS tokens, backdrop filters, and Three.js 3D WebGL backgrounds (Silk & MoltenMetal).

3. Team Roles & Contribution Breakdown

Jagadeesh Chinta (Lead Full-Stack AI Architect)	End-to-end full-stack development, Gemini Multimodal Vision API integration, prompt engineering, resilience cascading, offline heuristic compiler, live camera scanner, SSE code streaming, Liquid Glass design system, Vercel deployment, and evaluation benchmark suite.
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Note: This project was designed and built independently by Jagadeesh Chinta.

4. Key Features & Challenges Solved

Challenge 1: Resolving LLM Domain Bias
Problem: Initial vision prompts produced generic SaaS dashboards even when given restaurant menus or e-commerce wireframes.
Solution: Re-engineered prompts with strict OCR text extraction and domain-aware routing rules, ensuring sketches with "MENU", "HOURS", "LOCATION", or "PRICE" synthesize authentic restaurant websites with real dish cards and reservation forms.

Challenge 2: Robust Image Ingestion Across Formats
Problem: Data URLs in varying formats (UTF-8 SVGs, WebP canvas blobs) caused Gemini base64 decoding errors.
Solution: Engineered parseImageData in /api/analyze to handle Base64, raw SVG XML, and multipart data seamlessly.

Challenge 3: Vercel Serverless Function Timeouts
Problem: Deep multimodal code synthesis takes 15–20s, exceeding default Vercel function timeouts (10s).
Solution: Configured maxDuration = 60 and dynamic = "force-dynamic" across all Next.js App Router API routes.

5. Evaluation Benchmarks & Metrics

Evaluation Metric	Measured Score	Standard / Target	Status
Component Detection Recall	96.2%	> 85% required	PASSED
End-to-End Latency	8.4 seconds	< 15 seconds	PASSED
Average Token Cost	\$0.00018 / run	< \$0.01 / run	PASSED
Daily Cost (1,000 runs)	\$0.18 / day	< \$1.00 / day ceiling	PASSED
Offline Resilience Mode	100% Functional	Fallback Required	PASSED

6. Submission Links & Access Details

Live Production URL	https://sketch2-code-yd7c-git-master-jc-project.vercel.app
GitHub Repository	https://github.com/jagadeeshchinta/Sketch2Code
Evaluation Harness	https://sketch2-code-yd7c-git-master-jc-project.vercel.app/eval
Live Camera Studio	https://sketch2-code-yd7c-git-master-jc-project.vercel.app/create/live
Demo Video File	Included in this Google Drive folder (Demo_Video_WhiteboardOS.mp4)